

3.8.1 Centre Assessed Route T (PSA/ISA)

Each centre assessed unit comprises:

- Practical Skills Assessment (PSA)
- Investigative Skills Assignment (ISA).

The PSA consists of the centre's assessment of the candidate's ability to demonstrate practical skills throughout the course; thus, candidates should be encouraged to carry out practical and investigative work throughout the course of their study. This work should cover the skills and knowledge of How Science Works (Section 3.7) and in Sections 3.3 and 3.6.

The ISA has two stages where candidates:

- undertake practical work, collect and process of data
- complete a written ISA test.

There are two windows of assessment for the ISA:

- one for the practical work (Stage 1)
- one for the written test (Stage 2).

Each stage of the ISA must be carried out

- under controlled conditions
- within the windows of assessment stipulated by AQA in the Instructions for Administration of the ISA.

All students at a centre must complete the written test in a single uninterrupted session on the same day.

The ISA is set externally by AQA, but internally marked, with marking guidelines provided by AQA. In a given academic year two ISAs at each of AS and A2 level will be provided.

Practical Skills Assessment (PSA)

Candidates are assessed throughout the course on practical skills, using a scale from 0-9. The mark submitted for practical skills should be judged by the teacher. Teachers may wish to use this section for formative assessment and should keep an ongoing record of each candidate's performance but the mark submitted should represent the candidate's practical abilities over the whole course. Please refer to section 3.8.3 for marking guidance and criteria.

The nature of the assessment

Since the skills in this section involve implementation they must be assessed while the candidate is carrying out practical work. Practical activities are not intended to be undertaken as formal tests and supervisors can provide the usual level of guidance that would normally be given during teaching. In order to provide appropriate opportunities to demonstrate the necessary skills, instructions provided must not be too prescriptive but should allow candidates to make decisions for themselves, particularly concerning the conduct of practical work, their organisation and the manner in which equipment is used.

The tasks

There are no specific tasks set by AQA in relation to the PSA. Centres should set up tasks in order for the candidates to be provided opportunities to use the equipment deemed appropriate at the given level. Further guidance can be provided by the Assessment Adviser attached to the centre. Details of the appropriateness of the equipment and techniques are provided in Unit 3 and Unit 6 (Section 3.3 and 3.6).

The assessment criteria

In the context of material specified in the relevant AS or A2 specification candidates will be assessed on the following skills:

- Following instructions
- Selecting and using equipment
- Organisation and safety.

Detailed descriptors for these three skills are provided in Section 3.8.3.

AQA may wish to ask for further supporting evidence from centres in relation to the marks awarded for the PSA. Centres should therefore keep records of their candidates' performances in their practical activities throughout the course. (For example, a laboratory diary, log or tick sheet.)

Further guidance for awarding of marks for the PSA will be provided in the Teaching and learning resources web page.

Use of ICT during PSA

Candidates are encouraged to use ICT where appropriate in the course of developing practical skills, for example in collecting and analysing data.

Investigative Skills Assignment (ISA)

The Investigative Skills Assignment carries 41 marks and has two stages.

Stage 1: Collection and Processing of data

Candidates carry out practical work following an AQA task sheet. Centres may use the task sheet, as described, or may make minor suitable modifications to materials or equipment following AQA guidelines. Any modifications made to the task sheet must be agreed in writing with the AQA Assessment Adviser. The task may be conducted in a normal timetabled lesson but must be under controlled conditions and during the window of assessment for practical work.

Candidates will be asked to collect data and represent it in a table of their own design. They will be instructed to process the data and draw an appropriate graph. The teacher must not instruct the candidates on the presentation of the data or on the choice of graph/chart. All the completed work must be handed to the teacher at the end of the session. The teacher assesses the candidates' work to AQA marking guidelines.

There is no specified time limit for this stage.

Stage 2: The ISA written test

The ISA test should be taken after completion of Stage 1, under controlled conditions and during the window of assessment for the written test. All students at a centre must complete the written test in a single uninterrupted session on the same day. Each candidate is provided with an ISA test and the candidate's completed material from Stage 1. The teacher uses the AQA marking guidelines to assess the ISA test.

The ISA test is in two Sections:

a) Section A

This consists of a number of questions relating to the candidate's own data.

b) Section B

This section will provide a further set of data related to the original experiment. A number of questions relating to analysis and evaluation of the data then follow.

The number of marks allocated to each section may vary slightly with each ISA test.

Use of ICT during ISA

ICT may be used during the ISA Stages 1 and 2 but teachers should note any restrictions in the ISA marking guidelines. Use of the internet is not permitted.

Candidates absent for the practical work

A candidate absent for the practical work (Stage 1) should be given an opportunity to carry out the practical work before they sit the ISA test. This may be with another group or at a different time. In extreme circumstances when such arrangements are not possible, the teacher can supply a candidate with class data. In this case candidates cannot be awarded marks for Stage 1, but can still be awarded marks for Stage 2 of the assessment.

Material from AQA

For each ISA, AQA will provide:

- Teachers' Notes
- Task sheet
- ISA written test
- Marking guidelines.

This material must be kept under secure conditions within the centre. The centre must ensure security of the materials.

Further details regarding this material will be provided.

Security of assignments

All ISA materials including marked ISAs should be treated like examination papers and kept under secure conditions until the publication of results.

General Information

Route T

Administration

In any year a candidate may attempt either or both of the two ISAs. AQA will stipulate windows of assessment during which the ISAs (task and test) must be completed.

For each candidate, the teacher should submit to AQA a total mark comprising:

- The PSA mark
- The better ISA mark (if two have been attempted).

The ISA component of this mark must come from one ISA only, i.e. the marks awarded for individual stages of different ISAs cannot be combined.

The total mark must be submitted to AQA by the due date in the academic year for which the ISA was published.

Candidates may make only one attempt at an ISA and redrafting is not permitted at any stage during the ISA.

Work to be submitted

For each candidate in the sample the following materials must be submitted to the moderator by the deadline issued by AQA:

- the candidate's data from Stage 1
- the ISA written test which includes the Candidate Record Form, showing the marks for the ISA and the PSA.

In addition each centre must provide:

- a Centre Declaration Sheet

- details of any agreed amendments to the task sheet, with information supporting the changes from the AQA Assessment Adviser.

Working in groups

For the PSA candidates may work in groups provided that any skills being assessed are the work of individual candidates. For the ISA further guidance will be provided in the Teacher Notes.

Other information

Section 6 of this specification outlines further guidance on the supervision and authentication of centre assessed units.

Section 6 also provides information in relation to the internal standardisation of marking for these units. Please note that the marking of both of the PSA and the ISA must be internally standardised, as stated in Section 6.4.

Further support

AQA support the centre assessed units in a number of ways:

- AQA hold annual standardising meetings on a regional basis for all internally assessed components. Section 6 of this specification provides further details about these meetings
- Teaching and learning resources web page which includes information and guidance
- Assessment Advisers are appointed by AQA to provide advice on centre assessed units. Every centre is allocated an Adviser. Details are sent to the Head of Department.

The assessment advisers can provide guidance on:

- issues relating to the carrying out of assignments for assessment
- application of the marking guidelines.

Any amendments to the ISA task sheet must be discussed with the Assessment Adviser and confirmation of the amendments made must be submitted to the AQA moderator.